

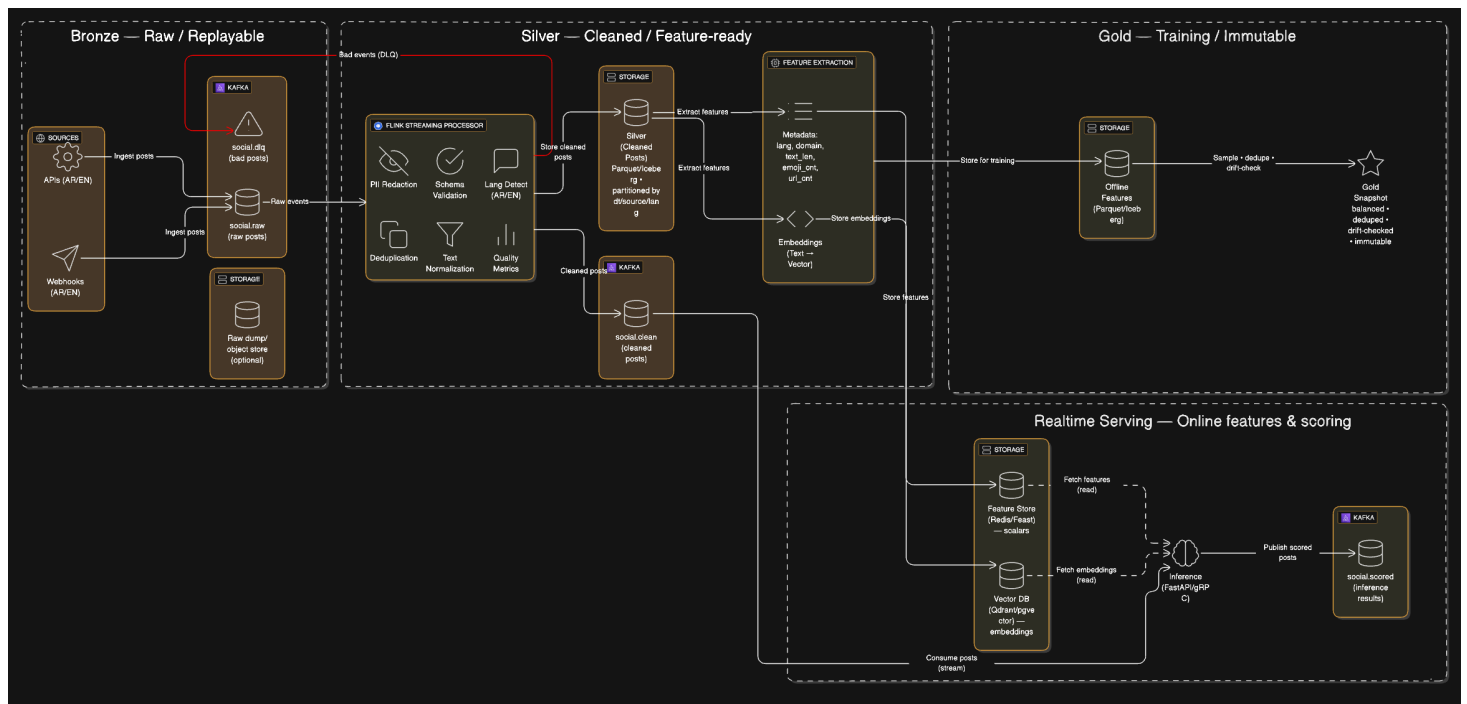
Real-Time Social Media Sentiment Pipeline

Overview

This project demonstrates a **data pipeline prototype** that prepares raw social media posts (Arabic/English) for:

- **Model training** (historical datasets, offline feature store, gold snapshots).
- **Real-time inference** (online feature store/vector DB, scoring service).

Architecture



The pipeline follows a **Bronze → Silver → Gold + Realtime Serving** pattern:

- **Bronze (Raw / Replayable)**
 - Ingests posts from APIs and Webhooks.
 - Stores into Kafka (social.raw) and optional object store dump.
 - Bad events sent to DLQ (social.dlq).

- **Silver (Cleaned / Feature-ready)**
 - Flink Streaming Processor applies:
 - PII redaction
 - Schema validation
 - Language detection
 - Deduplication
 - Text normalization
 - Quality metrics
 - Output stored in:
 - Kafka topic (`social.clean`)
 - Silver Parquet table (partitioned by dt/source/lang).
- **Gold (Training / Immutable)**
 - Feature extraction generates metadata + embeddings.
 - Stored in offline Parquet/Iceberg.
 - Periodic gold snapshot sampling → balanced, deduped, drift-checked dataset for model training.
- **Realtime Serving (Online features & scoring)**
 - Features in Redis (scalars) and Vector DB (pgvector).
 - Inference service fetches features + embeddings, consumes posts, and publishes scored outputs to Kafka (`social.scored`).

Prototype

The prototype simulates the architecture in **Python** with lightweight components:

- **Bronze Layer:**
 - `bronze_ingest.py` → loads `sample_posts.json` → produces into `social.raw.jsonl`.
 - DLQ handling for malformed/null records.
- **Silver Layer:**
 - `silver_clean.py` → regex/heuristics for PII redaction, normalization, deduplication, lang detect.
 - Output into `silver/social.clean.jsonl`.
 - Metrics & offsets tracked for replayability.
- **Feature Extraction:**
 - `silver_features.py` → simple hashing vector + metadata (`lang, domain, len, emoji_cnt, url_cnt`).

- **Gold Layer:**

Raw Ingest (Bronze)

```

1 {"post_id": "p101", "text": "I love the new phone update! Battery life is amazing 📶", "lang_hint": "en", "author_id": "u77", "created_at": "2025-09-25T12:15:00Z", "source": "x", "domain": "tech"}
2 {"post_id": "p202", "text": "التجربة كانت سيئة جدًا تواصلوا معي: 123-456-789 email@domain.com", "lang_hint": "ar", "author_id": "u88", "created_at": "2025-09-25T13:00:00Z", "source": "ig", "domain": "customer_service"}
3 {"post_id": "p203", "text": "Great service!!! thanks @support contact me at john.doe@mail.com", "lang_hint": "en", "author_id": "u15", "created_at": "2025-09-25T11:00:00Z", "source": "fb", "domain": "customer_service"}
4 {"post_id": "p204", "text": "تحديث الكاميرا ممتاز جدًا!!!! 📸 رابط: www.example.com", "lang_hint": "ar", "author_id": "u91", "created_at": "2025-09-25T09:41:00Z", "source": "x", "domain": "tech"}
5 {"post_id": "p205", "text": "Battery life still poor after the update :( ", "lang_hint": "en", "author_id": "u77", "created_at": "2025-09-25T12:25:00Z", "source": "x", "domain": "tech"}
6 {"post_id": "p205", "text": "Battery life still poor after the update :( ", "lang_hint": "en", "author_id": "u77", "created_at": "2025-09-25T12:25:00Z", "source": "x", "domain": "tech"}
7 {"post_id": "p206", "text": "Just had an amazing experience with customer support! Resolved my issue quickly 🌟", "lang_hint": "en", "author_id": "u99", "created_at": "2025-09-25T14:30:00Z", "source": "ig", "domain": "customer_service"}
8 {"post_id": "p207", "text": "إبراز: التحديث الجديد سيء للغاية، البطارية تفقد سرعة. رابط: www.badupdate.com", "lang_hint": "ar", "author_id": "u100", "created_at": "2025-09-25T15:00:00Z", "source": "fb", "domain": "tech"}
9 {"post_id": "p208", "text": "Loving the camera improvements in the latest update! 📷 #photography", "lang_hint": "en", "author_id": "u77", "created_at": "2025-09-25T16:00:00Z", "source": "x", "domain": "tech"}
10

```

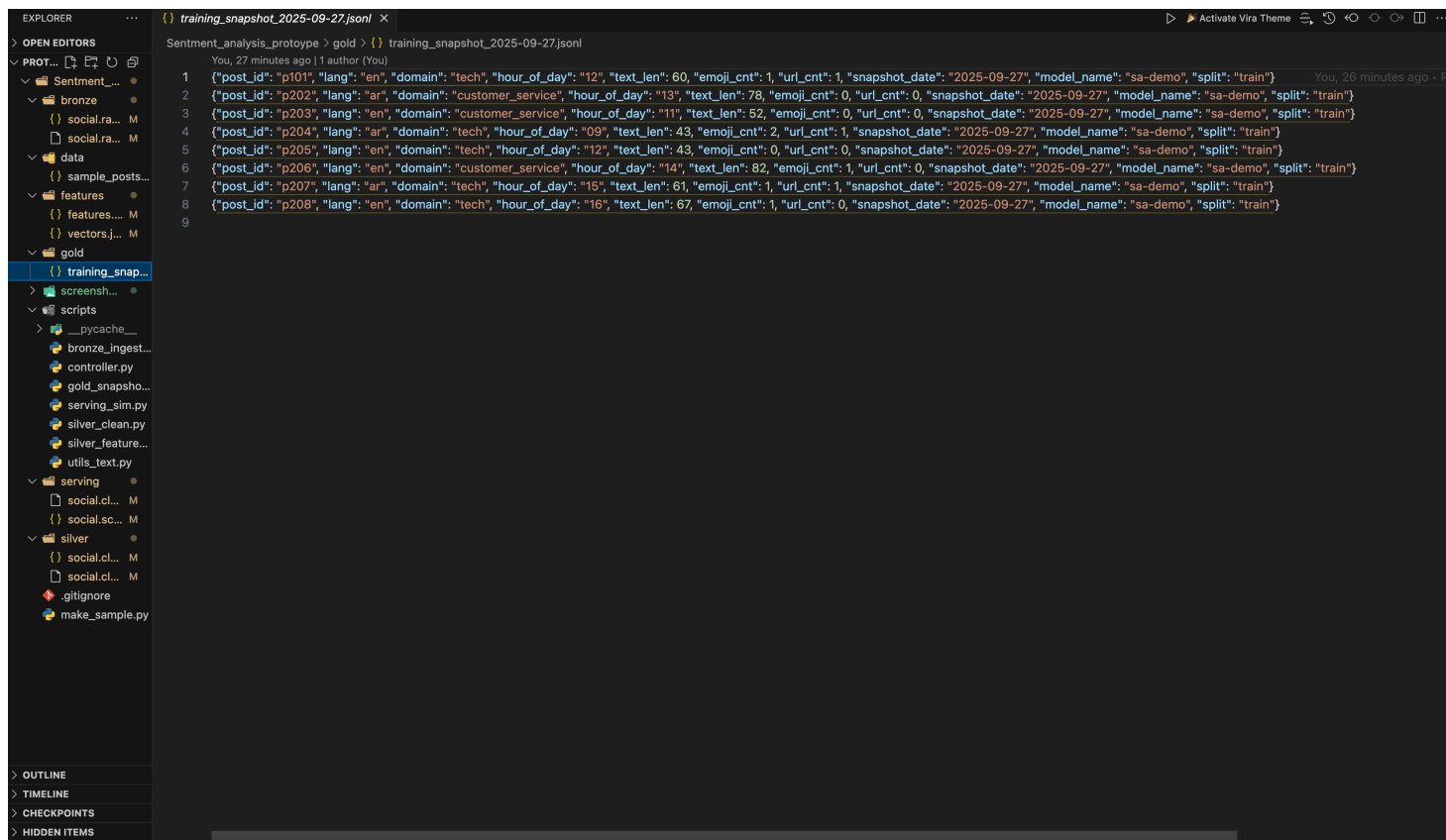
Silver Cleaned Data & Offsets

```
{ } sample_posts.json { } social.raw.json M { } social.raw.json.offset M { } social.clean.json M X
Sentiment_analysis_prototype > silver > { } social.clean.jsonl
You, 22 seconds ago | 1 author (You)
1 ('post_id': 'p101', 'source': 'x', 'domain': 'tech', 'created_at': '2025-09-25T12:15:00Z', 'lang': 'en', 'clean_text': 'i love the new phone update! battery life is amazing 🚀<url>',
2     'record_hash': 'b05d25a50d27d8a7', 'pii_found': 0, 'emoji_cnt': 1, 'url_cnt': 1, 'text_len': 60)
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4 ('post_id': 'p202', 'source': 'ig', 'domain': 'customer_service', 'created_at': '2025-09-25T13:00:00Z', 'lang': 'ar', 'clean_text': 'كرويت خدمة العملاء اليوم، التجربة كانت سيئة جدا. تواصلوا معي.<phone> او<em
5     'record_hash': '30b33026c6e6aefde', 'pii_found': 1, 'emoji_cnt': 0, 'url_cnt': 0, 'text_len': 78)
6
7 ('post_id': 'p203', 'source': 'fb', 'domain': 'customer_service', 'created_at': '2025-09-25T11:00:00Z', 'lang': 'en', 'clean_text': 'great service!! thanks <user> contact me at <email>',
8     'record_hash': '70f46ff39859c9fb2', 'pii_found': 1, 'emoji_cnt': 0, 'url_cnt': 0, 'text_len': 52)
9
10 ('post_id': 'p204', 'source': 'x', 'domain': 'tech', 'created_at': '2025-09-25T09:41:00Z', 'lang': 'ar', 'clean_text': 'رابط 📡 🔥 تحديث الكاميرا ممتاز جدا!!!!<url>',
11     'record_hash': 'e18ca0dcc59ccf40', 'pii_found': 0, 'emoji_cnt': 2, 'url_cnt': 1, 'text_len': 43)
12
13 ('post_id': 'p205', 'source': 'x', 'domain': 'tech', 'created_at': '2025-09-25T12:25:00Z', 'lang': 'en', 'clean_text': 'battery life still poor after the update. (:',
14     'record_hash': '1205b3712a9acebf', 'pii_found': 0, 'emoji_cnt': 0, 'url_cnt': 0, 'text_len': 43)
15
16 ('post_id': 'p206', 'source': 'ig', 'domain': 'customer_service', 'created_at': '2025-09-25T14:30:00Z', 'lang': 'en', 'clean_text': 'just had an amazing experience with customer support! resolved my is
17     'record_hash': 'd58a7d5a88694c4e', 'pii_found': 0, 'emoji_cnt': 1, 'url_cnt': 0, 'text_len': 82)
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19 ('post_id': 'p207', 'source': 'fb', 'domain': 'tech', 'created_at': '2025-09-25T15:00:00Z', 'lang': 'ar', 'clean_text': 'رابط 📡 🔥 التحديث الجديد سوء للغاية، البطارية تفترغ بسرعة.<url>',
20     'record_hash': '6de123198e794d5b', 'pii_found': 0, 'emoji_cnt': 1, 'url_cnt': 1, 'text_len': 61)
21
22 ('post_id': 'p208', 'source': 'x', 'domain': 'tech', 'created_at': '2025-09-25T16:00:00Z', 'lang': 'en', 'clean_text': 'loving the camera improvements in the latest update! 📷#photography',
23     'record_hash': 'c98e9a4ccb7a6e7', 'pii_found': 0, 'emoji_cnt': 1, 'url_cnt': 0, 'text_len': 67)
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25 ('post_id': 'p101', 'source': 'x', 'domain': 'tech', 'created_at': '2025-09-25T12:15:00Z', 'lang': 'en', 'clean_text': 'i love the new phone update! battery life is amazing 🚀<url>',
26     'record_hash': 'b05d25a50d27d8a7', 'pii_found': 0, 'emoji_cnt': 1, 'url_cnt': 1, 'text_len': 60)
27
28 ('post_id': 'p202', 'source': 'ig', 'domain': 'customer_service', 'created_at': '2025-09-25T13:00:00Z', 'lang': 'ar', 'clean_text': 'كرويت خدمة العملاء اليوم، التجربة كانت سيئة جدا. تواصلوا معي.<phone> او<em
29     'record_hash': '30b33026c6e6aefde', 'pii_found': 1, 'emoji_cnt': 0, 'url_cnt': 0, 'text_len': 78)
30
31 ('post_id': 'p203', 'source': 'fb', 'domain': 'customer_service', 'created_at': '2025-09-25T11:00:00Z', 'lang': 'en', 'clean_text': 'great service!! thanks <user> contact me at <email>',
32     'record_hash': '70f46ff39859c9fb2', 'pii_found': 1, 'emoji_cnt': 0, 'url_cnt': 0, 'text_len': 52)
33
34 ('post_id': 'p204', 'source': 'x', 'domain': 'tech', 'created_at': '2025-09-25T09:41:00Z', 'lang': 'ar', 'clean_text': 'رابط 📡 🔥 تحديث الكاميرا ممتاز جدا!!!!<url>',
35     'record_hash': 'e18ca0dcc59ccf40', 'pii_found': 0, 'emoji_cnt': 2, 'url_cnt': 1, 'text_len': 43)
36
37 ('post_id': 'p205', 'source': 'x', 'domain': 'tech', 'created_at': '2025-09-25T12:25:00Z', 'lang': 'en', 'clean_text': 'battery life still poor after the update. (:',
38     'record_hash': '1205b3712a9acebf', 'pii_found': 0, 'emoji_cnt': 0, 'url_cnt': 0, 'text_len': 43)
39
40 ('post_id': 'p206', 'source': 'ig', 'domain': 'customer_service', 'created_at': '2025-09-25T14:30:00Z', 'lang': 'en', 'clean_text': 'just had an amazing experience with customer support! resolved my is
41     'record_hash': 'd58a7d5a88694c4e', 'pii_found': 0, 'emoji_cnt': 1, 'url_cnt': 0, 'text_len': 82)
42
43 ('post_id': 'p207', 'source': 'fb', 'domain': 'tech', 'created_at': '2025-09-25T15:00:00Z', 'lang': 'ar', 'clean_text': 'رابط 📡 🔥 التحديث الجديد سوء للغاية، البطارية تفترغ بسرعة.<url>',
44     'record_hash': '6de123198e794d5b', 'pii_found': 0, 'emoji_cnt': 1, 'url_cnt': 1, 'text_len': 61)
45
46 ('post_id': 'p208', 'source': 'x', 'domain': 'tech', 'created_at': '2025-09-25T16:00:00Z', 'lang': 'en', 'clean_text': 'loving the camera improvements in the latest update! 📷#photography',
47     'record_hash': 'c98e9a4ccb7a6e7', 'pii_found': 0, 'emoji_cnt': 0, 'url_cnt': 0, 'text_len': 67)
```

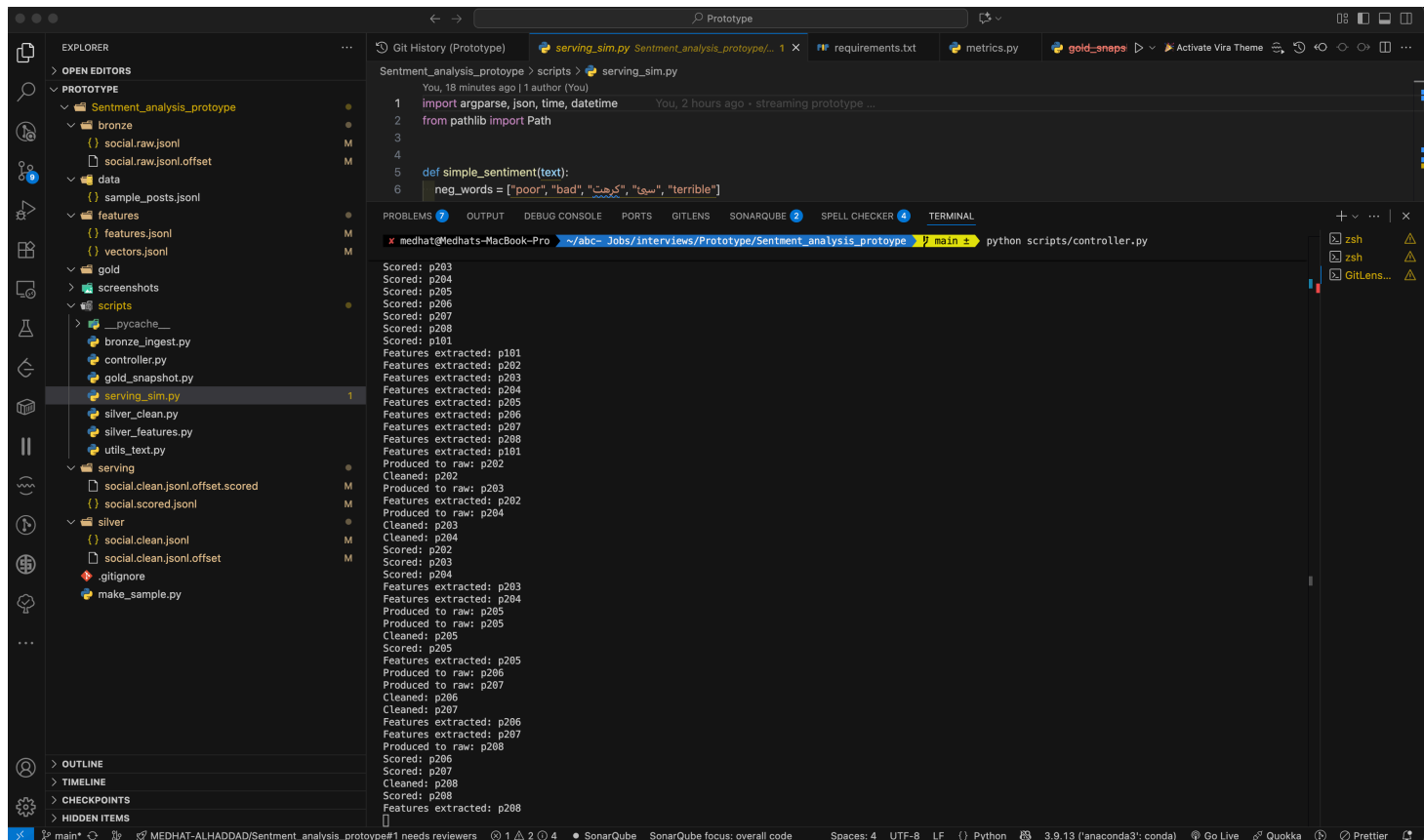
Features & Vectors

[illegible]

Gold Snapshot



Running Controller



Serving & Scored Output

```
{ } social.scored.jsonl M ×
Sentiment_analysis_prototype > serving { } social.scored.jsonl
You, 9 minutes ago | 1 author (You)

1  [{"post_id": "p101", "sentiment": "positive", "prob_neg": 0.1, "prob_neu": 0.1, "prob_pos": 0.8, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:06.403349"}
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4  [{"post_id": "p204", "sentiment": "positive", "prob_neg": 0.1, "prob_neu": 0.1, "prob_pos": 0.8, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:06.403606"}
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6  [{"post_id": "p206", "sentiment": "neutral", "prob_neg": 0.2, "prob_neu": 0.6, "prob_pos": 0.2, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:06.403627"}
7  [{"post_id": "p207", "sentiment": "neutral", "prob_neg": 0.2, "prob_neu": 0.6, "prob_pos": 0.2, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:06.403637"}
8  [{"post_id": "p208", "sentiment": "neutral", "prob_neg": 0.2, "prob_neu": 0.6, "prob_pos": 0.2, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:06.403647"}
9  [{"post_id": "p101", "sentiment": "positive", "prob_neg": 0.1, "prob_neu": 0.1, "prob_pos": 0.8, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:06.403860"}
10 [{"post_id": "p202", "sentiment": "negative", "prob_neg": 0.8, "prob_neu": 0.1, "prob_pos": 0.1, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:10.413701"}
11 [{"post_id": "p203", "sentiment": "neutral", "prob_neg": 0.2, "prob_neu": 0.6, "prob_pos": 0.2, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:10.413750"}
12 [{"post_id": "p204", "sentiment": "positive", "prob_neg": 0.1, "prob_neu": 0.1, "prob_pos": 0.8, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:10.413771"}
13 [{"post_id": "p205", "sentiment": "negative", "prob_neg": 0.8, "prob_neu": 0.1, "prob_pos": 0.1, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:12.416371"}
14 [{"post_id": "p206", "sentiment": "neutral", "prob_neg": 0.2, "prob_neu": 0.6, "prob_pos": 0.2, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:16.428416"}
15 [{"post_id": "p207", "sentiment": "neutral", "prob_neg": 0.2, "prob_neu": 0.6, "prob_pos": 0.2, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:16.428505"}
16 [{"post_id": "p208", "sentiment": "neutral", "prob_neg": 0.2, "prob_neu": 0.6, "prob_pos": 0.2, "model_version": "sa-demo-v1", "scored_at": "2025-09-27T09:01:16.429051"}
17
```

Quick Start

1. Clone the repo

```
git clone https://github.com/<your-org>/sentiment_pipeline_prototype.git
cd sentiment_pipeline_prototype
```

2. Run the pipeline controller

This orchestrates Bronze → Silver → Features → Serving.

```
python scripts/controller.py
```

You should see logs like:

Produced to raw: p101
Cleaned: p101
Features extracted: p101
Scored: p101

3. Check the outputs

After running the controller, the pipeline will generate files at each stage. You can inspect them directly:

- **Bronze (raw ingested posts):** `bronze/social.raw.jsonl` >> Contains the original posts exactly as ingested.
- **Silver (cleaned posts):** `silver/social.clean.jsonl` >> Posts after schema validation, PII redaction, normalization, deduplication, and language detection.
- **Features (metadata + vectors):** `features/features.jsonl` , `features/vectors.jsonl` >> Metadata features (language, domain, text length, emoji count, URL count) and hashed text embeddings.
- **Serving (scored posts):** `serving/social.scored.jsonl` >> Posts enriched with toy sentiment labels (positive/negative/neutral).
- **-Gold Snapshot (for training):** `gold/training_snapshot_<date>.json` >> Immutable, balanced, deduplicated dataset sampled from Silver for model training.

4. Schedule gold snapshot

To automate daily **Gold snapshot** creation, schedule `gold_snapshot.py` using cron (Linux/macOS) or Task Scheduler (Windows).

Example cron job (runs every day at 02:00 server time):

```
0 2 * * * cd /path/to/repo && python scripts/gold_snapshot.py >> logs/gold_snapshot.log
```

Strategic Discussion

Q1. How would you analyze the data and ensure quality for training?

- **Validation:** enforce schema checks, PII detection, null filtering in Silver.
- **Metrics:** track volume, deduplication %, language distribution, drift in features.
- **Drift Checks:** compare feature distributions between current Silver vs past snapshots.
- **Gold Snapshot Rules:** balanced across domains/lang, deduped by record hash, immutable storage for reproducibility.

Q2. How would you integrate with a feature store for training and inference?

- **Offline Store (Parquet/Iceberg):**
 - Historical features saved daily for training.
 - Used to build gold snapshots.
- **Online Store (Redis/Feast + Vector DB):**
 - Scalars → Redis/Feast for fast lookups.
 - Embeddings → Vector DB for similarity search.
 - Inference service queries both in real time.

This ensures **consistency** (same features for training & inference) and **low latency serving** for production AI workloads.